

Theorem 5.1:

$$b(x;n,\theta) = b(n-x;n,1-\theta)$$

Proof:

$$b(x;n,\theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x}$$

$$b(n-x;n,(1-\theta)) = \binom{n}{n-x} (1-\theta)^{n-x} \theta^{n-(n-x)}$$

$$\begin{aligned} \binom{n}{n-x} &= \frac{n!}{(n-x)! (n-(n-x))!} \\ &= \frac{n!}{(n-x)! x! \binom{n}{x}} \end{aligned}$$

$$b(x;n,\theta) = b(n-x;n,1-\theta)$$

When $n=10$ and $x=3$;

$$\begin{aligned} b(7; 10, 0.80) &= b(10 - 7; 10, 1 - 0.80) \\ &= b(3; 10, 0.20) \\ &= B(3; 10, 0.20) - B(2; 10, 0.20) \\ &= 0.8791 - 0.6778 \\ &= 0.2013 \end{aligned}$$

Theorem 5.2: The mean and the variance of the binomial distribution are,

$$\mu = n\theta$$
$$\sigma^2 = n\theta(1-\theta)$$

Proof:

$$\begin{aligned}\mu = E[X] &= \sum_{x=0}^n x \cdot \binom{n}{x} \theta^x (1-\theta)^{n-x} \\ &= \sum_{x=1}^n x \cdot \frac{n!}{x!(n-x)!} \theta^x (1-\theta)^{n-x} \\ &= \sum_{x=1}^n \frac{n!}{(x-1)!(n-x)!} \theta^x (1-\theta)^{n-x}\end{aligned}$$

Omit the term corresponding to $x=0$ which is zero.

$$\begin{aligned}\mu &= n\theta \sum_{x=1}^n \frac{(n-1)!}{(x-1)!(n-x)!} \theta^{x-1} (1-\theta)^{n-x} \\ \mu &= n\theta \sum_{x=1}^n \binom{n-1}{x-1} \theta^{x-1} (1-\theta)^{n-x}\end{aligned}$$

Let $y=x-1$, $m=n-1$

$$\mu = n\theta \sum_{y=0}^m \binom{m}{y} \theta^y (1-\theta)^{m-y} = n\theta$$

Sum of all values of a binomial distribution with parameters m and θ and were $=1$.

$$\sigma^2 = \mu'_2 - \mu^2$$

$$\text{Let } E[X^2] = E[X(X-1)] + E[X]$$

$$E[X(X-1)] = \sum_{x=0}^n x(x-1) \binom{n}{x} \theta^x (1-\theta)^{n-x}$$

$$= \sum_{x=2}^n \frac{n!}{(x-2)! (n-x)!} \theta^x (1-\theta)^{n-x}$$

$$= n(n-1)\theta^2 \sum_{x=2}^n \frac{(n-2)!}{(x-2)! (n-x)!} \theta^{x-2} (1-\theta)^{n-x}$$

$$= n(n-1)\theta^2 \sum_{x=2}^n \binom{n-2}{x-2} \theta^{x-2} (1-\theta)^{n-x}$$

Let $y=x-2$ and $m=n-2$

$$E[X(X-1)] = n(n-1)\theta^2 \sum_{y=0}^m \binom{m}{y} \theta^y (1-\theta)^{m-y}$$

$$= n(n-1)\theta^2$$

$$\mu_2' = E[X(X-1)] - E[X]^2 = n(n-1)\theta^2 + n\theta$$

$$\sigma^2 = \mu_2' - \mu^2$$

$$= n(n-1)\theta^2 + n\theta - n^2\theta^2$$

$$= n^2\theta^2 - n\theta^2 + n\theta - n^2\theta^2$$

$$= n\theta(1-\theta)$$

Reading the Binomial Table

[illegible]

5. $\mathcal{L}$

Theorem 5.4: Moment generating function of the binomial distribution is given by,

$$M_x(t) = [1+\theta(e^t-1)]^n$$

Proof:

By definition

$$\begin{aligned} M_x(t) &= E[e^{xt}] = \sum_{x=0}^n e^{xt} \binom{n}{x} \theta^x (1-\theta)^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (e^t)^x (1-\theta)^{n-x} \end{aligned}$$

The summation can be easily recognized as the binomial expansion of $[\theta e^t + (1-\theta)]^n$

$$M_x(t) = [1+\theta(e^t-1)]^n$$

Use above moment generating function to verify the result of theorem 5.2.

$$\begin{aligned} M'_x(t) &= n[1+\theta(e^t-1)]^{n-1} \cdot \theta e^t \\ &= n\theta e^t [1+\theta(e^t-1)]^{n-1} \end{aligned}$$

Upon substituting $t=0$ we get,

$$\mu_1' = n\theta e^0 [1 + \theta(e^0 - 1)]^{n-1}$$

$$\mu_1' = n\theta$$

$$\begin{aligned} M_x''(t) &= n\theta e^t [1 + \theta(e^t - 1)]^{n-1} \\ &\quad + n(n-1)\theta^2 e^{2t} [1 + \theta(e^t - 1)]^{n-2} \\ &= n\theta e^t [1 + \theta(e^t - 1)]^{n-2} (1 + \theta(e^t - 1) \\ &\quad + (n-1)\theta e^t) \\ &= n\theta e^t (1 + \theta e^t - \theta + n\theta e^t \\ &\quad - \theta e^t) [1 + \theta(e^t - 1)]^{n-2} \\ &= n\theta e^t (1 - \theta + n\theta e^t) [1 + \theta(e^t - 1)]^{n-2} \end{aligned}$$

$$\begin{aligned} \mu_2' &= n\theta e^0 (1 - \theta + n\theta e^0) [1 + \theta(e^0 - 1)]^{n-2} \\ &= n\theta (1 - \theta + n\theta) \end{aligned}$$

$$\begin{aligned} \sigma^2 &= \mu_2' - \mu^2 \\ &= n\theta (1 - \theta + n\theta) - n^2 \theta^2 \\ &= n\theta - n\theta^2 + n^2 \theta^2 - n^2 \theta^2 \\ \sigma^2 &= n\theta (1 - \theta) \end{aligned}$$

4. Negative Binomial Distribution

In connection with repeated Bernoulli trials, we are sometimes interested in the number of the trials on which the k^{th} success occurs.

Example:

The probability that 5th trial will be the third head from flipping a balanced coin.

Probability, that a burglar will be caught for the second time on his eighth job.

If the k^{th} success occurs in x^{th} trial, there must be $k-1$ successes on the first $x-1$ trials. Therefore probability is,

$$b(k-1; x-1, \theta) = \binom{x-1}{k-1} \theta^{k-1} (1-\theta)^{x-k}$$

Probability of a success on the x^{th} trial is θ . Then the probability that the k^{th} success occurs on the x^{th} trial is,

$$\theta \cdot b(k-1; x-1, \theta) = \binom{x-1}{k-1} \theta^k (1-\theta)^{x-k}$$

Definition 5.4: A random variable x has a negative binomial distribution and it is referred as a negative binomial random variable if and only if,

$$b^*(x; k, \theta) = \binom{x-1}{k-1} \theta^k (1-\theta)^{x-k} \text{ for } x=k, k+1, k+2, \dots$$

Note: Thus the number of trials on which the k^{th} success occurs is a random variable having a negative binomial distribution with the parameters k and θ .

Example: If the probability is 0.40 that a child exposed to a certain contagious disease will catch it. What is the probability that the tenth child exposed to the disease will be the third to catch it?

$$\begin{aligned} b^*(10;3,0.40) &= \binom{10-1}{3-1} (0.4)^3 (1-0.4)^{10-3} \\ &= \binom{10-1}{3-1} (0.4)^3 (0.6)^7 \\ &= 0.0645 \end{aligned}$$

When a table of binomial probabilities is available, the determination of negative binomial probabilities can generally be simplified by making use of the following identity.

Theorem 5.5:

$$b^*(x;k,\theta) = \frac{k}{x} b(k;x,\theta)$$

Proof:

$$\begin{aligned} b^*(x;k,\theta) &= \binom{x-1}{k-1} \theta^k (1-\theta)^{x-k} \\ &= \frac{(x-1)!}{(k-1)! (x-k)!} \theta^k (1-\theta)^{x-k} \end{aligned}$$

$$\begin{aligned}
&= \frac{k}{x} \cdot \frac{x!}{k!(x-k)!} \theta^k (1-\theta)^{x-k} \\
&= \frac{k}{x} \cdot b(k;x,\theta)
\end{aligned}$$

Since negative binomial distribution with $k=1$ has many important application, It is given a special name **“Geometric distribution”**.

Definition 5.5:

A random variable X has a geometric distribution and it is referred to as a geometric random variable if and only if its probability distribution is given by,

$$g(x;\theta) = \theta(1-\theta)^{x-1} \text{ for } x=1,2,3,\dots$$

Example: If the probability is 0.75 that an applicant for a driver's license will pass the road on any given try, what is the probability that an applicant will finally pass the test on the fourth try?

$$X=4, \theta=0.75$$

$$\begin{aligned}g(4;0.75) &= 0.75(1-0.75)^{4-1} \\ &= 0.75(0.25)^3 \\ &= 0.0117\end{aligned}$$

Theorem 5.6:

The mean and the variance of the negative binomial distribution are,

$$\mu = \frac{k}{\theta} \text{ and } \sigma^2 = \frac{k}{\theta} \left(\frac{1}{\theta} - 1 \right)$$